E. Packing and minimum dimension (proof of Proposition 3)

In this section, we elaborate on the geometric capacity question introduced in the main text: how large must the embedding dimension d be for linear probes to realize all concept combinations in $\mathcal{C}$.

We use hyperplane-arrangement counting bounds to formalize this. Intuitively, each concept contributes n decision boundaries, and to classify all combinations these boundaries must carve enough regions in $\mathbb{R}^d$ to accommodate all n^k combinations. We first recall the relevant region-count results, then apply them to prove the lower bound $d \geq k$.

For $d \geq 1$, we work with affine hyperplanes (the setting used by linear probes with bias):

$$H_{\boldsymbol{w},b} = \{\, \boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{w}^\intercal \boldsymbol{x} + b = 0 \,\}, \qquad \boldsymbol{w} \neq 0,\ b \in \mathbb{R}.$$

An *arrangement* $\mathcal{H} = \{H_1, \ldots, H_m\}$ is a family of hyperplanes. It is in *general position* when no $d+1$ hyperplanes intersect at a common point, which maximizes the number of connected regions in $\mathbb{R}^d$ (Zaslavsky, 1975; Ziegler, 1995).

This allows us to count the number of connected regions in $\mathbb{R}^d$ that an arrangement of m hyperplanes carves out, a classical result that we state below.

Theorem 1 (Zaslavsky's region bounds in general position (Zaslavsky, 1975))**.** Let $\mathcal{H}$ be an arrangement of m hyperplanes in $\mathbb{R}^d$ that is in general position. Then, the number of connected regions $R(\mathcal{H})$ is given by:

(a) *Affine (with a bias) case.* If the hyperplanes may carry arbitrary offsets b_i (so $\mathcal{H}$ is not required to be central), then

$$R(\mathcal{H}) = R_{\text{aff}}(m, d) := \sum_{r=0}^{d} \binom{m}{r}.$$

(b) *Central case.* If every hyperplane passes through the origin,

$$R(\mathcal{H}) = R_{\text{lin}}(m, d) := 2 \sum_{r=0}^{d-1} \binom{m-1}{r}.$$

Note that for $d < k$, one has

$$R_{\text{aff}}(k, d) = \sum_{r=0}^{d} \binom{k}{r} < 2^k,$$

which is the key inequality we will need. We now exploit Theorem 1 to prove the lower bound on probe dimension; first for the binary case, then for general n.

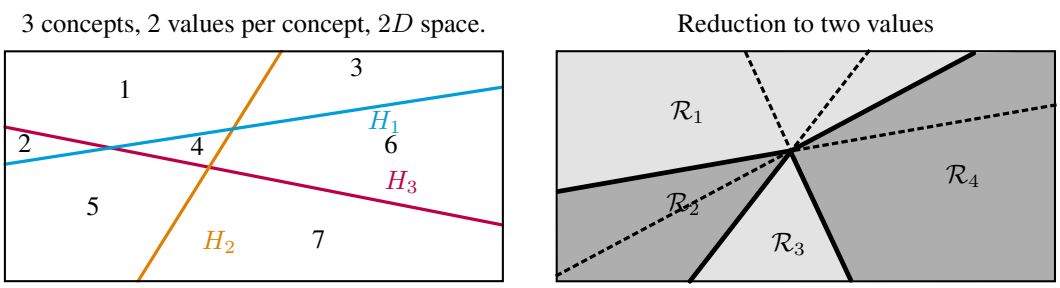

Figure 15. **Illustration of the probe dimension lower bound.** Schematic of probe hyperplane arrangements and induced regions in embedding space. **(left)** For $k = 3$ in $d = 2$, three affine hyperplanes partition the plane into at most 7 regions, fewer than the $2^3 = 8$ binary concept combinations; therefore $d \geq 3$. **(right)** For $n > 2$, restricting each concept to any two values induces a binary subproblem. Hence any model that realizes all n^k combinations must also realize these binary restrictions, so the binary lower bound applies to the non-binary case as well.

Proposition 3 (Minimum dimension for linear probes)**.** For k concepts, each with n values, suppose there exist linear probes that correctly classify each concept value for all n^k combinations from embeddings $f(\boldsymbol{x}) \in \mathbb{R}^d$. Then necessarily $d \geq k$.

Proof sketch. Reduce to the binary case by fixing two values per concept and restricting to the resulting 2^k combinations. Each concept induces one affine separating hyperplane. To realize all binary labelings, the arrangement must carve at least 2^k regions. By Theorem 1, when $d < k$ we have $\sum_{r=0}^{d} \binom{k}{r} < 2^k$, so 2^k regions are impossible. Hence $d \geq k$. Tightness follows by a $d = k$ construction (coordinates per concept with suitable affine offsets). See Fig. 15.

Proof. First, we "reduce" the problem to the binary case. For each concept $i \in [k]$, take two distinct $a_i, b_i \in [n]$ and restrict the label space to

$$\mathcal{C}^{(2)} := \prod_{i=1}^{k} \{a_i, b_i\}, \qquad |\mathcal{C}^{(2)}| = 2^k.$$

Since every label combination in $[n]^k$ is classified correctly, all tuples in $\mathcal{C}^{(2)}$ are classified as well. For each concept i, we can define the induced binary score on only the considered values

$$h_i^{(a_i,b_i)}(\boldsymbol{z}) := h_{i,a_i}(\boldsymbol{z}) - h_{i,b_i}(\boldsymbol{z}) = (\boldsymbol{w}_{i,a_i} - \boldsymbol{w}_{i,b_i})^\intercal \boldsymbol{z} + (b_{i,a_i} - b_{i,b_i}). \tag{62}$$

Let $\boldsymbol{w}_i := \boldsymbol{w}_{i,a_i} - \boldsymbol{w}_{i,b_i}$ and $\beta_i := b_{i,a_i} - b_{i,b_i}$. Each induced binary classifier defines an affine hyperplane

$$\mathcal{H}_i := \{\boldsymbol{z} \in \mathbb{R}^d \mid \boldsymbol{w}_i^\intercal \boldsymbol{z} + \beta_i = 0\}. \tag{63}$$

Because multiclass predictions are correct on $\mathcal{C}^{(2)}$, each concept i is correctly separated between values a_i and b_i on this restricted set. Hence the k affine hyperplanes $\mathcal{H}_1, \ldots, \mathcal{H}_k$ must separate $\mathbb{R}^d$ into at least 2^k distinct regions.

But the number of regions formed by k affine hyperplanes in $\mathbb{R}^d$ is at most

$$\sum_{r=0}^{d} \binom{k}{r} < 2^k \quad \text{whenever } d < k \quad \text{(by Theorem 1)}. \tag{64}$$

Thus, we must have $d \geq k$.

Construction is simple. Let $d = k$, and assume (for $\boldsymbol{e}_i \in \mathbb{R}^d$ with $\boldsymbol{e}_{ij} = \delta_{ij}$)

$$\boldsymbol{z}_{\boldsymbol{c}} := \sum_{i}^{k} \boldsymbol{e}_i c_i, \tag{65}$$

i.e. each embedding is a sum of standard basis unit vectors for each concept scaled by the value of that concept.

Then we can define probe vectors as

$$\boldsymbol{w}_{i,j} := 2j\, \boldsymbol{e}_i \quad \text{and} \quad b_{i,j} := -j^2. \tag{66}$$

Then

$$h_{i,j}(\boldsymbol{z}_{\boldsymbol{c}}) = 2j\boldsymbol{e}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}} - j^2 = 2c_i - j^2 = c_i^2 - (j - c_i)^2. \tag{67}$$

Therefore, for each i, $\arg\max_{j \in [n]} h_{i,j}(\boldsymbol{z}_{\boldsymbol{q}}) = q_i$, so all n^k combinations are correctly classified in $d = k$. □

For intuition of the constructed case, see Section F.1.

F. Examples of linearly compositional models and their geometries

We give two toy geometries that make probe behavior explicit under strong linear-factorization assumptions. They can be understood as lying on two extremes of the space of linear representations: a case of "perfect LRH" geometry, where all the factors of concept values are co-linear (Section F.1), and a case where factors are implied to be orthogonal to any other factor, leading to a maximum possible dimensionality of linearly factorized spaces (Section F.2). These are *not* claims of full compositional generalization (they do not model the learning algorithm or stability), but provide intuition for probe geometry and dimensionality, especially in the cases where the classification behavior is intuitively desirable (Section F.2).

F.1. Case 1: Ideal "LRH" concept classifier

To gain intuition into the geometry of features and linear probes, we consider a commonly-assumed setting of the Linear Representation Hypothesis (LRH) (Elhage et al., 2022; Park et al., 2023; Rajendran et al., 2024), and consider the problem of finding probes capable of classifying the representations. Instead of a joint optimization, we assume the representations are already given and follow a strict linear-factorization structure under LRH.

Specifically, we make the following assumptions: (1) The representation for any input with concept tuple $\boldsymbol{c} = (c_1, \ldots, c_k) \in [n]^k$ is

$$\boldsymbol{z}_{\boldsymbol{c}} = \sum_{i=1}^{k} \alpha_{i,c_i} \boldsymbol{d}_i \quad \boldsymbol{d}_i \in \mathbb{R}^d. \tag{68}$$

(2) The concept direction vectors $\{\boldsymbol{d}_i\}_{i=1}^k \subset \mathbb{R}^d$ are fixed and orthogonal, i.e., $\boldsymbol{d}_i^\intercal \boldsymbol{d}_\ell = 0$ for $i \neq \ell$ (hence linearly independent and $d \geq k$)[3].

(3) For each concept i, its n values correspond to a known ordered set of scalar coefficients $\{\alpha_{i,j}\}_{j=1}^n$, with distinct values j for different i. Under these assumptions, orthogonality of the directions implies the representations form a regular grid in $\boldsymbol{z}$ inside V (each coordinate axis is one concept).

We consider the problem of finding the linear probes $\{\boldsymbol{w}_{i,j}\}$ that classify the representations:

$$\min_{\{\boldsymbol{w}_{i,j}, b_{i,j}\}} \quad \sum_{\boldsymbol{c}} \sum_{i=1}^{k} \left(-\log \frac{\exp\left(\boldsymbol{w}_{i,c_i}^\intercal \boldsymbol{z}_{\boldsymbol{c}} + b_{i,c_i}\right)}{\sum_{j=1}^n \exp\left(\boldsymbol{w}_{i,j}^\intercal \boldsymbol{z}_{\boldsymbol{c}} + b_{i,j}\right)} \right). \tag{69}$$

We consider the nearest-neighbor classifier. Concretely, we define the approximated concept-value prototypes by averaging representations over all tuples with concept i fixed to value j:

$$\widetilde{\boldsymbol{d}}_{i,j} := \frac{1}{n^{k-1}} \sum_{\boldsymbol{c} \in [n]^k : c_i = j} \boldsymbol{z}_{\boldsymbol{c}} = \alpha_{i,j} \boldsymbol{d}_i + \sum_{\ell \neq i} \bar{\alpha}_\ell \boldsymbol{d}_\ell := \alpha_{i,j} \boldsymbol{d}_i + \boldsymbol{m}_i, \tag{70}$$

where $\bar{\alpha}_\ell := \frac{1}{n} \sum_{r=1}^n \alpha_{\ell,r}$ and $\boldsymbol{m}_i := \sum_{\ell \neq i} \bar{\alpha}_\ell \boldsymbol{d}_\ell$. Using these prototypes, we set the corresponding affine probes as[4]

$$\widetilde{\boldsymbol{w}}_{i,j} = 2\widetilde{\boldsymbol{d}}_{i,j}, \qquad \widetilde{b}_{i,j} = -\|\widetilde{\boldsymbol{d}}_{i,j}\|_2^2. \tag{71}$$

We can verify that such a classifier works. The score for the j-th probe of concept i on an input $\boldsymbol{z}_{\boldsymbol{c}}$ (where the true value for concept i is c_i) is:

$$\begin{aligned} h_{i,j}(\boldsymbol{z}_{\boldsymbol{c}}) &:= \widetilde{\boldsymbol{w}}_{i,j}^\intercal \boldsymbol{z}_{\boldsymbol{c}} + \widetilde{b}_{i,j} \\ &= 2\widetilde{\boldsymbol{d}}_{i,j}^\intercal \boldsymbol{z}_{\boldsymbol{c}} - \|\widetilde{\boldsymbol{d}}_{i,j}\|_2^2 \\ &= 2(\alpha_{i,j}\boldsymbol{d}_i + \boldsymbol{m}_i)^\intercal \boldsymbol{z}_{\boldsymbol{c}} - \|\alpha_{i,j}\boldsymbol{d}_i + \boldsymbol{m}_i\|_2^2 \\ &= \left(2\alpha_{i,j}\boldsymbol{d}_i^\intercal(\boldsymbol{z}_{\boldsymbol{c}} - \boldsymbol{m}_i) - \alpha_{i,j}^2\|\boldsymbol{d}_i\|_2^2\right) + \left(2\boldsymbol{m}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}} - \|\boldsymbol{m}_i\|_2^2\right). \end{aligned}$$

[3] We consider this for simplicity; in general one may consider the directions to be linearly independent instead

[4] We find these weights by considering the power diagram (Aurenhammer, 1987) over the points $\widetilde{\boldsymbol{d}}_{i,j}$, the weights are biases are then remaps from the power diagram to the joint argmax classifier.